SMS 2101: GRAPH THEORY

Unless stated otherwise, all graphs are finite, simple, and undirected.
For a graph G, take $n = |V(G)|$ and $m = |E(G)|$.

I. Foundations

1. Let G have vertex set $\{1, 2, \dots, 7\}$ and edge set

$$E(G) = \{12, 13, 14, 23, 25, 35, 36, 45, 47, 56, 67\}.$$

   (a) Find the order, size, degree of each vertex, and degree sequence of G.

   (b) Let H have vertex set $\{1, 2, 3, 5\}$ and edge set $\{12, 13, 23, 25\}$. Is H a subgraph of G? Is it induced?

   (c) Give a spanning tree of G.

   (d) Find the degree sequence and size of $\overline{G}$ without listing all its edges.

2. Decide whether each description is possible. Justify each answer.

   (a) A graph has degree sequence $(6, 5, 4, 4, 3, 2, 2, 2)$.

   (b) At a meeting of eleven people, each person shakes hands with exactly five others.

   (c) A graph of order 9 has one vertex of each degree $0, 1, \dots, 8$.

3. For the graph G in Problem 1, determine whether each sequence is a walk, trail, path, closed walk, or cycle, and find its length.

$$W_1 = (1, 2, 3, 5, 2, 1), \quad W_2 = (7, 4, 5, 3, 6, 7), \quad W_3 = (2, 1, 3, 2, 5, 6, 3).$$

4. Apply the Havel-Hakimi algorithm to each vector below. State whether it is graphical. If it is, construct a graph having that degree sequence.

$$(5, 4, 4, 3, 3, 3, 2), \qquad (4, 4, 3, 1, 1, 1).$$

   Record every reduced vector used by the algorithm.

5. For the graph G in Problem 1:

   (a) find all geodesics from vertex 2 to vertex 7;

(b) find the eccentricity of every vertex; and

(c) determine $\mathrm{rad}(G)$, $\mathrm{diam}(G)$, and the centre of G.

6. Let

$$E(G_1) = \{12, 23, 34, 45, 51, 13\},$$

$$E(G_2) = \{ce, ea, ad, db, bc, ca\}.$$

(a) Give an isomorphism from G_1 to G_2.

(b) The graphs C_6 and $2K_3$ have the same degree sequence. Explain why they are not isomorphic.

7. Let Q have vertex set

$$\{a, b, c, x, y, z, w\}$$

and edge set

$$\{ax, ay, by, bz, cx, cz, cw\}.$$

(a) Find a bipartition of Q.

(b) Is this bipartition unique, apart from interchanging its two parts?

(c) Add the edge ac. Exhibit an odd cycle in the resulting graph.

8. For each statement below, decide whether the information guarantees that G is a tree. Prove your answer or give a counterexample.

(a) G is connected, has 12 vertices, and has 11 edges.

(b) G is acyclic, has 12 vertices, and has 11 edges.

(c) G has 12 vertices and 11 edges.

9. Let T be the tree with

$$V(T) = \{a, b, c, d, e, f, g, h, i\}, \qquad E(T) = \{ab, bc, cd, de, cf, cg, gh, hi\}.$$

Find its pendant vertices, diameter, radius, and centre. Also show the successive stages of the leaf-deletion procedure used to find the centre.

II. Core techniques

10. For each of K_5, $K_{2,4}$, and $K_{3,3}$, decide whether it is Eulerian and whether it is Hamiltonian. Whenever the answer is yes, exhibit an Eulerian circuit or a Hamiltonian cycle. Whenever it is no, give a reason.

11. Let R have vertices $v_1, \dots, v_5$ and edges

$$e_1 = v_1v_2, \; e_2 = v_1v_3, \; e_3 = v_2v_3, \; e_4 = v_2v_4, \; e_5 = v_3v_5, \; e_6 = v_4v_5.$$

   (a) Write the adjacency matrix $A(R)$ in the vertex order $v_1, \dots, v_5$.

   (b) Write the vertex-edge incidence matrix $B(R)$ using the edge order $e_1, \dots, e_6$.

   (c) Verify directly that $BB^T = A + D$, where D is the diagonal matrix of vertex degrees.

12. Use powers of the adjacency matrix from Problem 11 to find:

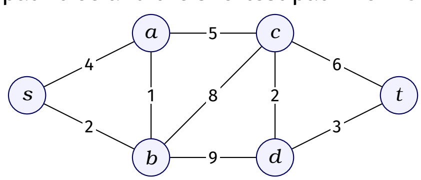
   (a) the number of walks of length 2 from v_1 to v_5;

   (b) the number of closed walks of length 3 starting at v_1; and

   (c) the total number of triangles in R.

   Explain how each answer is read from the appropriate matrix entry or trace.

13. Apply Dijkstra's algorithm to the weighted graph below, starting at s. At every iteration, record the permanent vertex, all tentative distances, and the predecessor of each unsettled vertex. Give the shortest-path tree and the shortest path from s to t.



   How do the shortest path and distance from s to t change if the weight of ct is changed from 6 to 2?

14. Let G be a graph of order n.

   (a) Express $\deg_{\overline{G}}(v)$ in terms of $\deg_G(v)$.

   (b) Prove that $A(\overline{G}) = J - I - A(G)$.

   (c) If G and $\overline{G}$ are both k-regular, determine n in terms of k, and show that k must be even.

III. Proof and synthesis

15. Prove the Handshaking Lemma. Deduce that:

   (a) every graph has an even number of vertices of odd degree; and

   (b) a k-regular graph of odd regularity has even order.

16. Prove that every graph on at least two vertices has two vertices of the same degree.

17. Prove that every graph on six vertices contains a triangle either in the graph or in its complement. Show that five vertices do not suffice by considering C_5. Thus establish that $R(3,3) = 6$.

18. Prove that if G is disconnected, then $\overline{G}$ is connected. In fact, show that any two vertices of $\overline{G}$ are at distance at most 2.

19. Let G be connected. Prove that if $\operatorname{diam}(G) \geq 3$, then $\operatorname{diam}(\overline{G}) \leq 3$.

20. Suppose G has order $n \geq 2$ and

$$\delta(G) \geq \frac{n-1}{2}.$$

Prove that G is connected. Then prove the stronger conclusion $\operatorname{diam}(G) \leq 2$ (and hence, in particular, $\operatorname{diam}(G) \leq 3$).

21. Prove that a graph is bipartite if and only if it contains no odd cycle. For the converse, fix a vertex in each component and partition the component according to whether distance from that vertex is even or odd.

22. Prove that a graph T is a tree if and only if there is a unique path between every pair of its vertices. Use this to prove that every tree of order n has exactly $n-1$ edges.

23. Prove that every non-trivial tree has at least two pendant vertices. Then justify the leaf-deletion method: deleting all pendant vertices from a tree with at least three vertices produces a smaller tree with the same centre. Deduce that the centre of a tree consists of either one vertex or two adjacent vertices.

24. (a) Prove that a self-complementary graph has order $4r$ or $4r+1$ for some non-negative integer r.
    (b) Verify explicitly that P_4 is self-complementary.
    (c) Let G be self-complementary. Form H from the disjoint union of G and a path $a-b-c-d$ by joining every vertex of G to b and c. Prove that H is self-complementary. Use this construction to give a self-complementary graph of order 8.

25. Prove that every vertex of an Eulerian graph has even degree. Give an example of a graph in which every vertex has even degree but which is not Eulerian, and explain which part of the definition fails.